

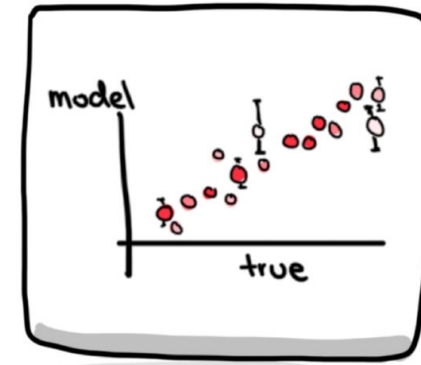
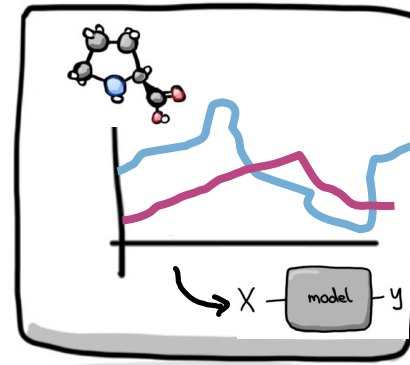
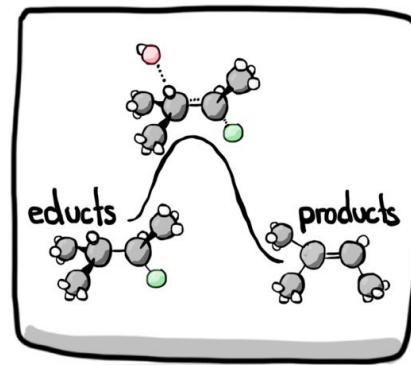
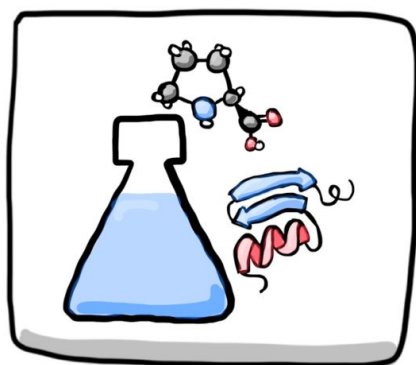
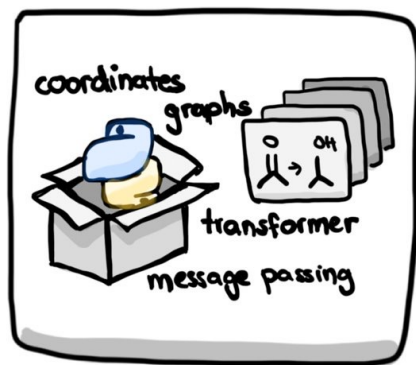


ML Predictions with ChemProp and ChemTorch

Ass. Prof. Esther Heid

Welcome to the Heid Lab ...

... where our **vision** is to establish a **virtual lab to understand + predict reactivity**



Johannes
Karwounopoulos



Jasper
De Landsheere



Leonard
Galustian



Leon
Ganser



Konstantin
Mark



Charlotte
Gerhaher



Maximilian
Kovar



Jonathan
Zheng



Anton
Zamyatin



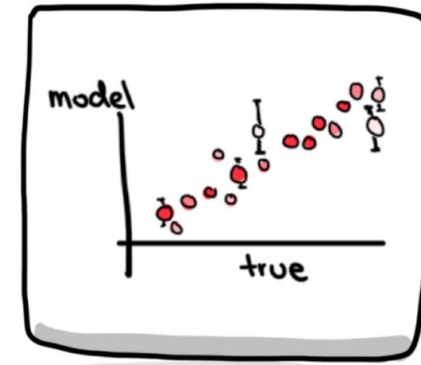
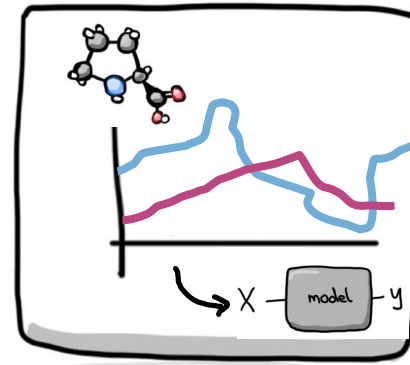
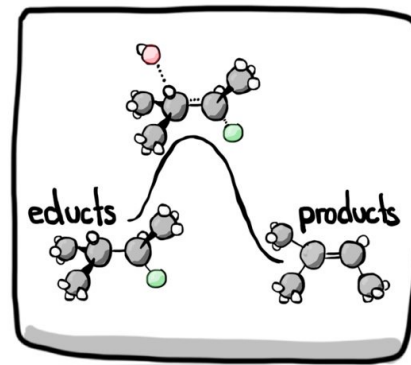
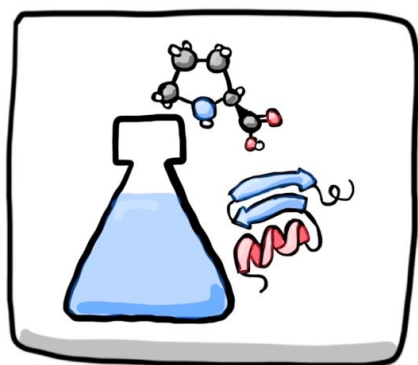
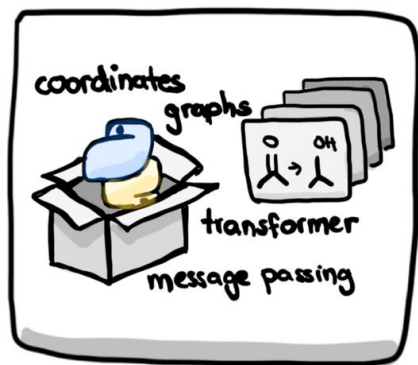
Synnöve
Lemke



Sebastian
Pinter

Welcome to the Heid Lab ...

... where our **vision** is to establish a **virtual lab to understand + predict reactivity**



Johannes
Karwounopoulos



Jasper
De Landsheere



Leonard
Galustian



Leon
Ganser



Konstantin
Mark



Charlotte
Gerhaher



Maximilian
Kovar



Jonathan
Zheng



Anton
Zamyatin



Synnöve
Lemke



Sebastian
Pinter

Machine learning for chemistry

What is a molecule?

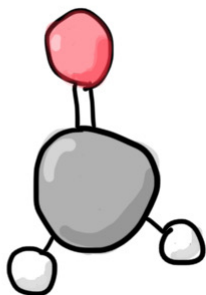
- Group of atoms (nuclei + electrons) which form dynamic structures and different conformers, and require a quantum-mechanical description

Machine learning for chemistry

What is a molecule?

- Group of atoms (nuclei + electrons) which form dynamic structures and different conformers, and require a quantum-mechanical description

1
1
0
0
1
0



C	0.00	0.00	0.00
O	1.15	-0.16	-0.20
H	-0.70	-0.83	0.12
H	-0.42	1.01	0.07

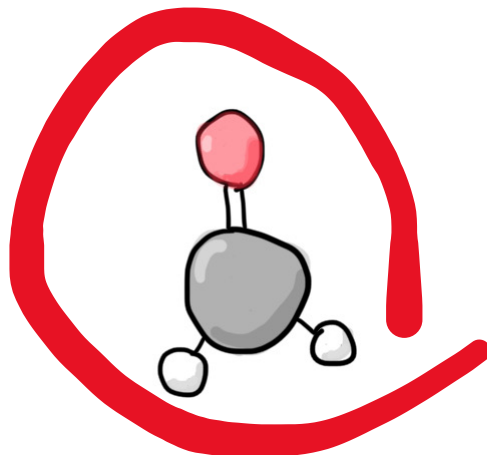
Choosing a **chemically meaningful representation and architecture** is essential for tailoring machine learning to chemistry

Machine learning for chemistry

What is a molecule?

- Group of atoms (nuclei + electrons) which form dynamic structures and different conformers, and require a quantum-mechanical description

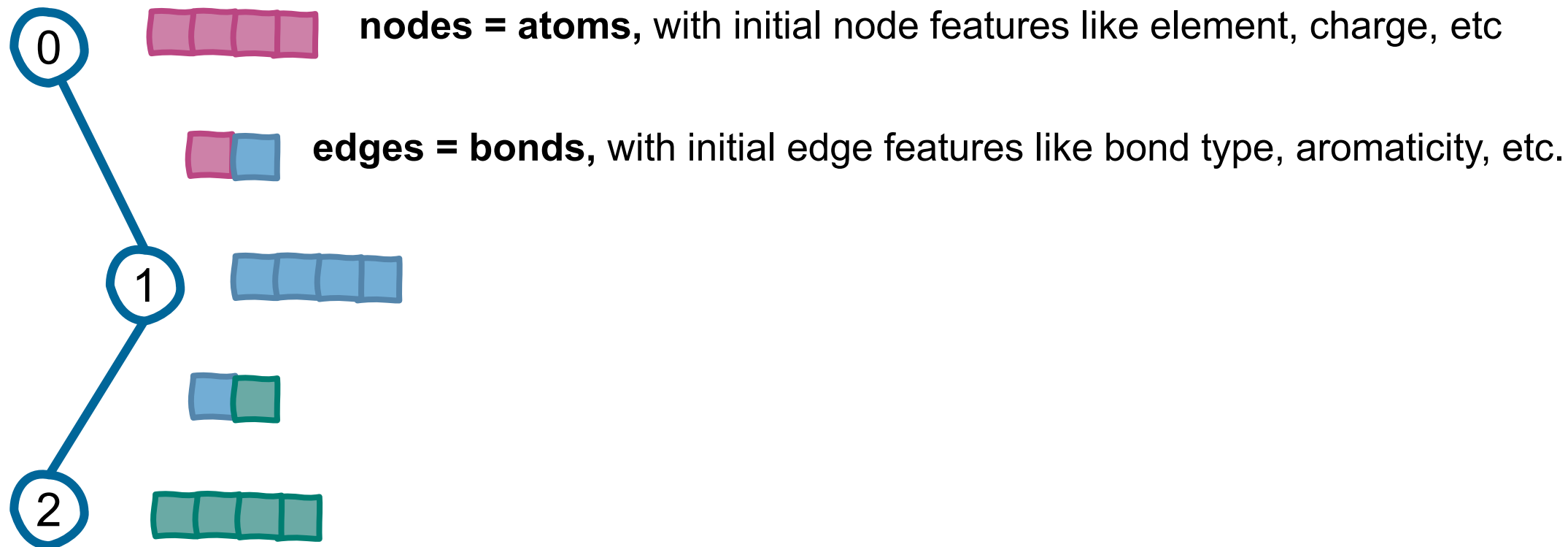
1
1
0
0
1
0



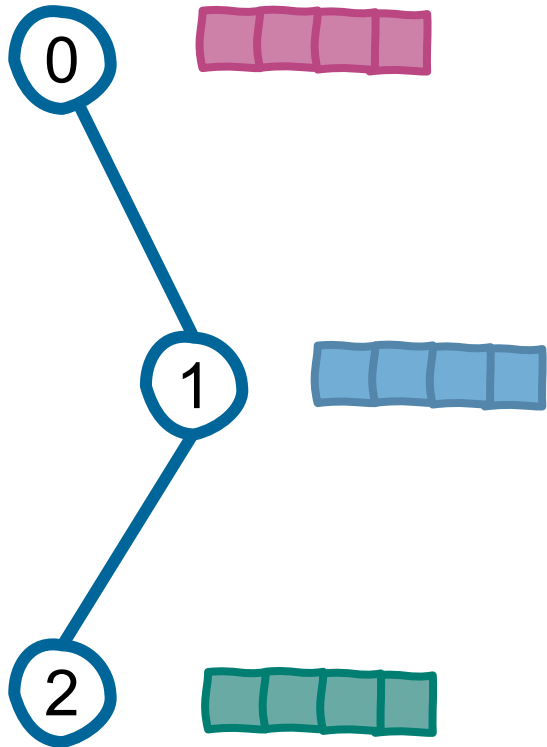
C	0.00	0.00	0.00
O	1.15	-0.16	-0.20
H	-0.70	-0.83	0.12
H	-0.42	1.01	0.07

Choosing a **chemically meaningful representation and architecture** is essential for tailoring machine learning to chemistry

Molecular graph

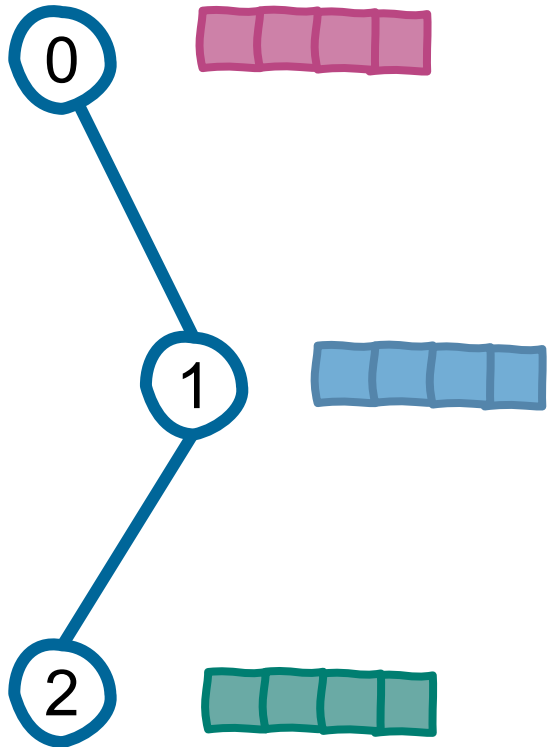


Graph convolutions



New vector  =  + NN()

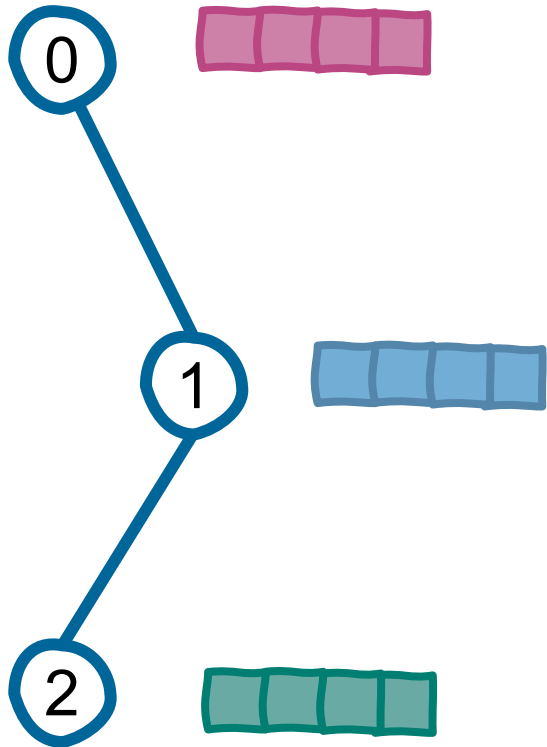
Graph convolutions



New vector  =  + NN()

New vector  =  + NN() + NN()

Graph convolutions

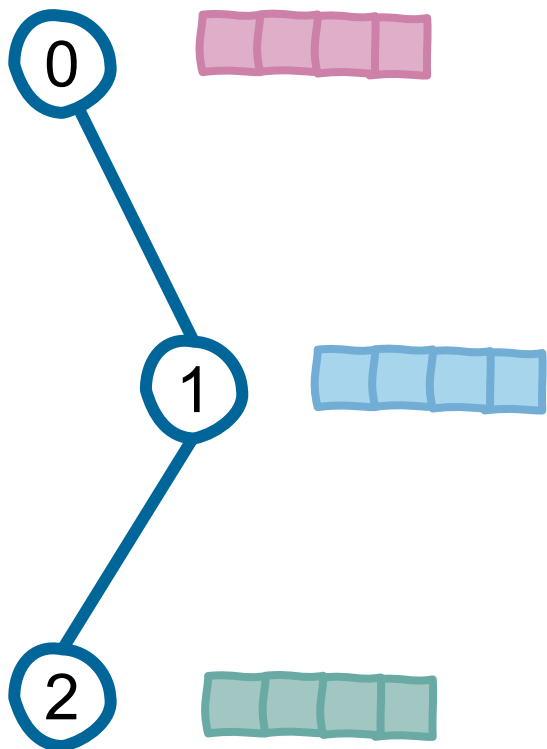


New vector  =  + NN()

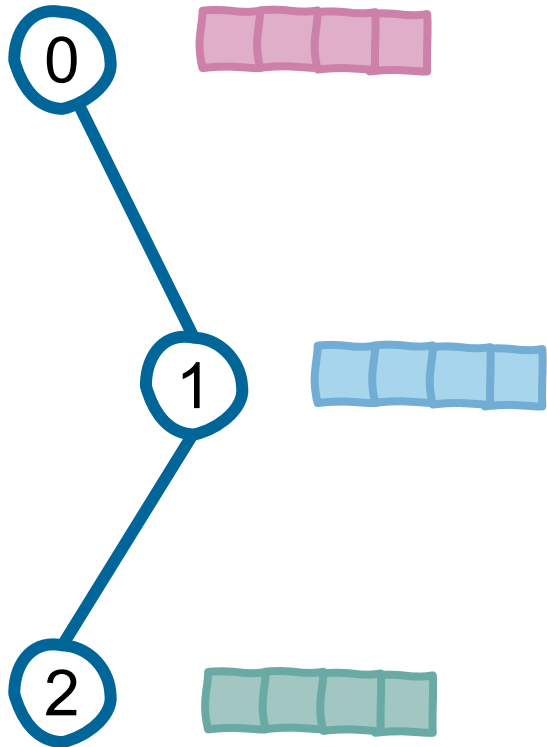
New vector  =  + NN() + NN()

New vector  =  + NN()

Graph convolutions

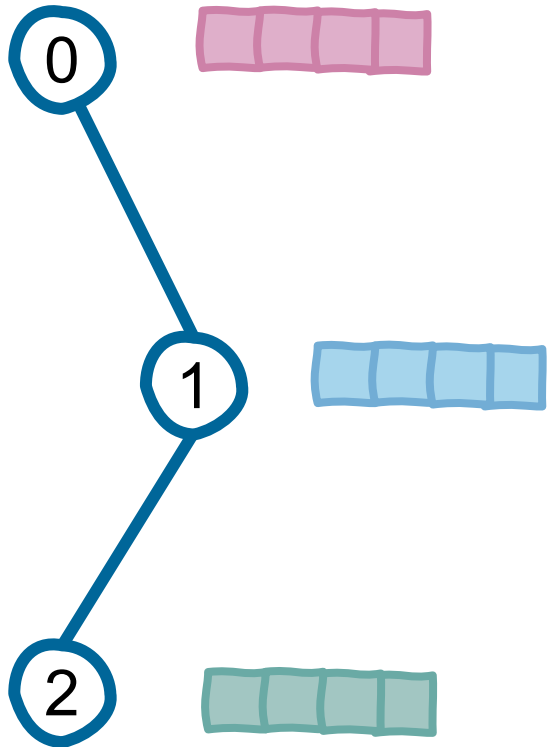


Graph convolutions



New vector  =  + NN()

Graph convolutions

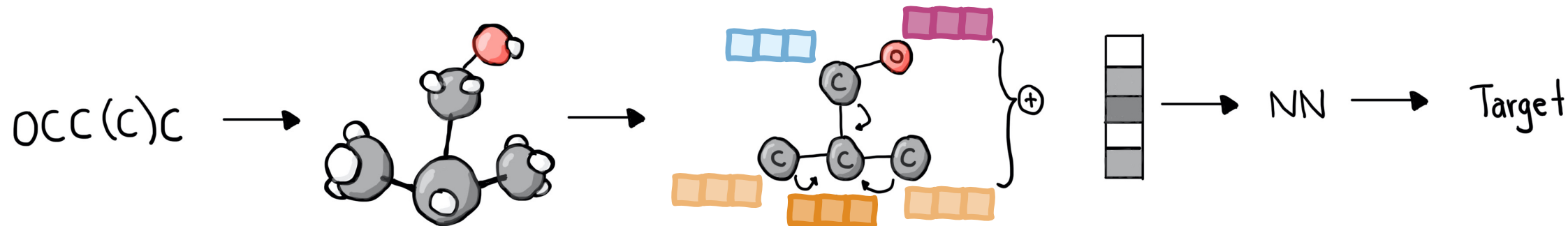


New vector $\begin{bmatrix} \end{bmatrix} = \begin{bmatrix} \end{bmatrix} + \text{NN}(\begin{bmatrix} \end{bmatrix})$

$= \left(\begin{bmatrix} \end{bmatrix} + \text{NN}(\begin{bmatrix} \end{bmatrix}) \right) + \text{NN}(\begin{bmatrix} \end{bmatrix} + \text{NN}(\begin{bmatrix} \end{bmatrix}) + \text{NN}(\begin{bmatrix} \end{bmatrix}))$

Molecular property prediction

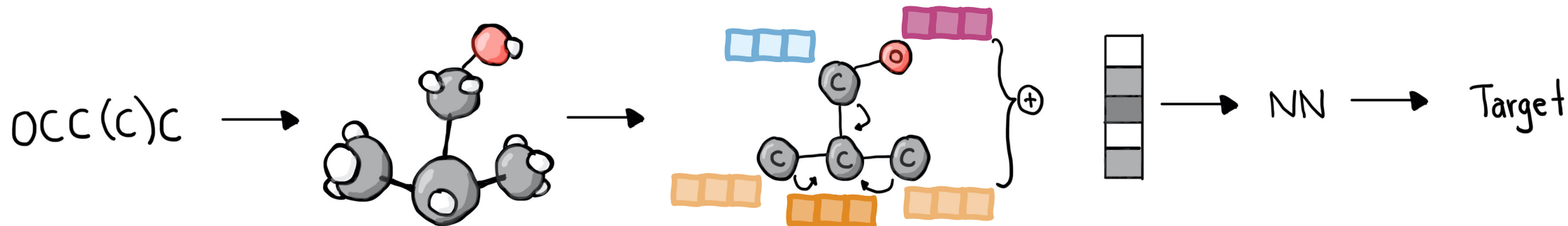
Learned fingerprints via graph-convolutional neural networks



Heid, Greenman, Chung, Li, Graff, Vermeire, Wu, Green. J. Chem. Inf. Model (2023), 64, 9

Molecular property prediction

Learned fingerprints via graph-convolutional neural networks



Example: Prediction of water/octanol partition coefficients (current state-of-the-art)

	SAMPL6	SAMPL7	SAMPL9
# Submissions	105	36	18
Average error (RMSE)	1.51	1.44	2.11
Lowest error (RMSE)	0.38	0.49	1.12
Ours	0.27	0.49	1.11

Heid, Greenman, Chung, Li, Graff, Vermeire, Wu, Green. J. Chem. Inf. Model (2023), 64, 9

Deployed in MIT's  chemprop

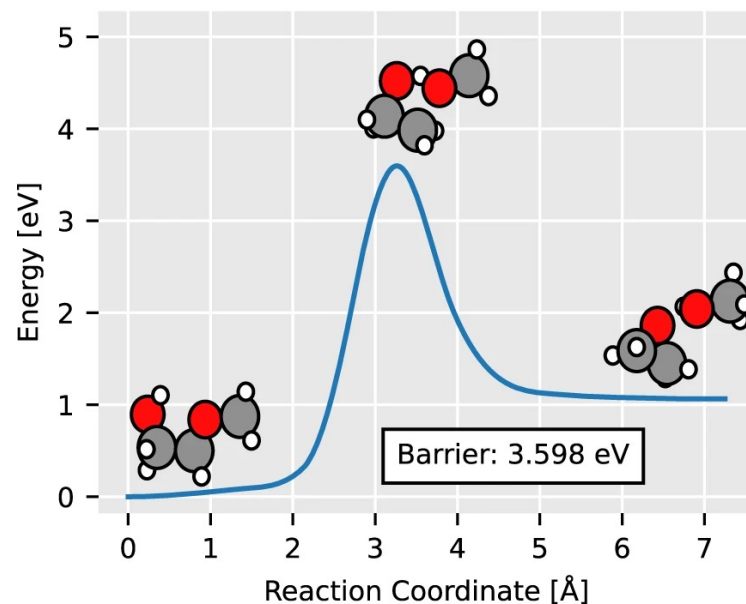
Hands-on Demo



chemprop

Part B-1

Reaction property prediction



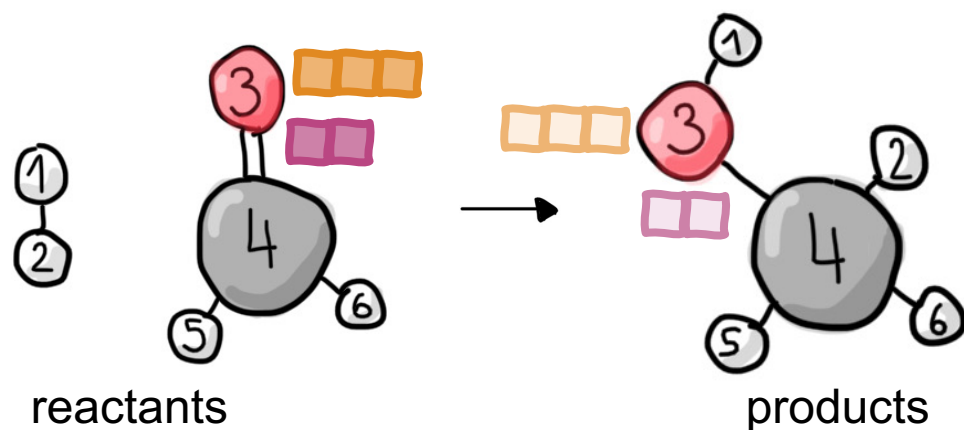
© Winther and coworkers, Sci. Data. 2022

Heid, Greenman, Chung, Li, Graff, Vermeire, Wu, Green. J. Chem. Inf. Model (2023), 64, 9

Heid, Green. J. Chem. Inf. Model (2022), 62, 2101

Reaction property prediction

Change molecular representation, keep architecture: **CGR**

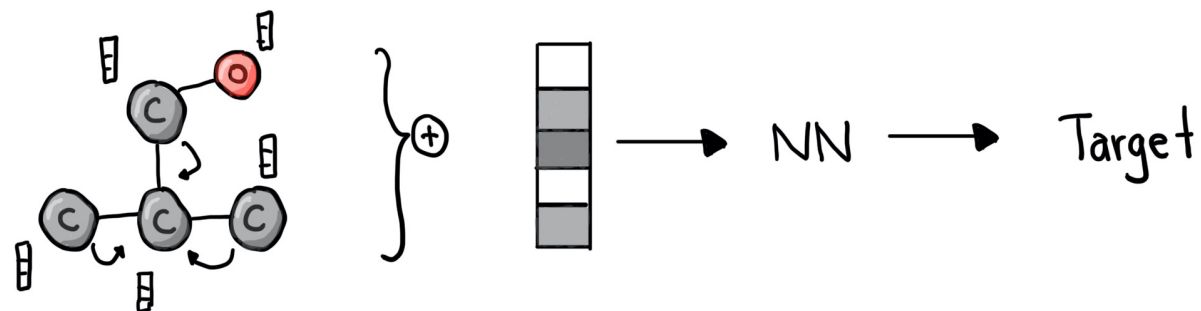
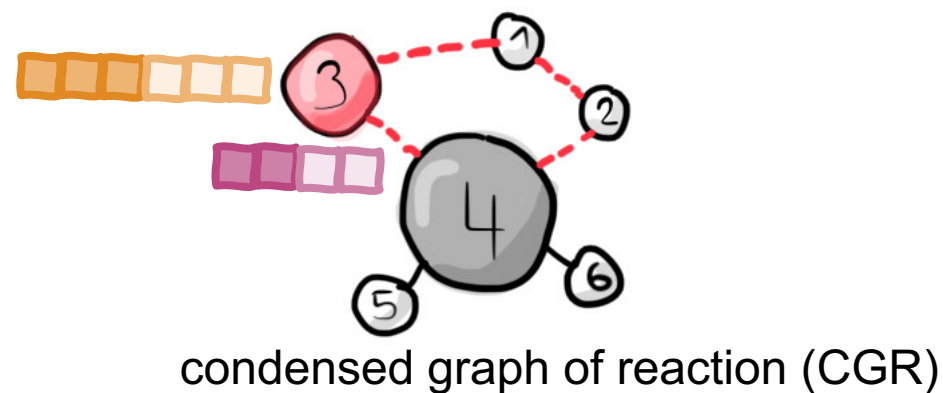
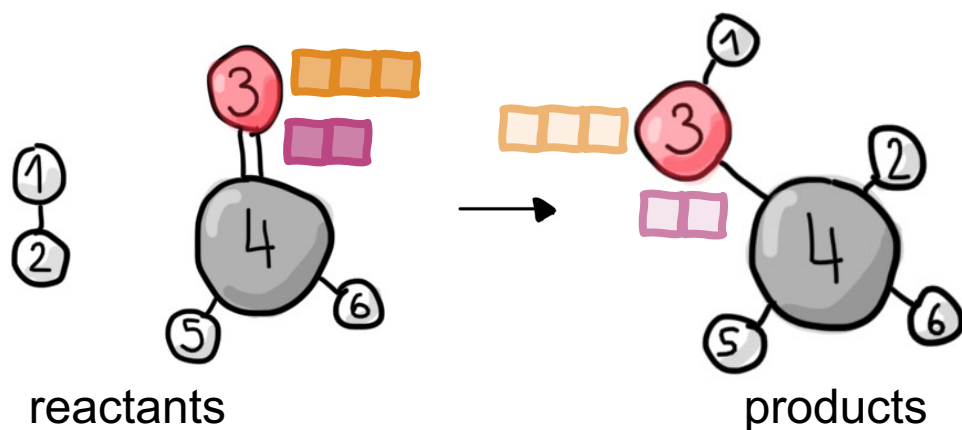


Heid, Greenman, Chung, Li, Graff, Vermeire, Wu, Green. J. Chem. Inf. Model (2023), 64, 9

Heid, Green. J. Chem. Inf. Model (2022), 62, 2101

Reaction property prediction

Change molecular representation, keep architecture: **CGR**



Heid, Greenman, Chung, Li, Graff, Vermeire, Wu, Green. J. Chem. Inf. Model (2023), 64, 9
Heid, Green. J. Chem. Inf. Model (2022), 62, 2101



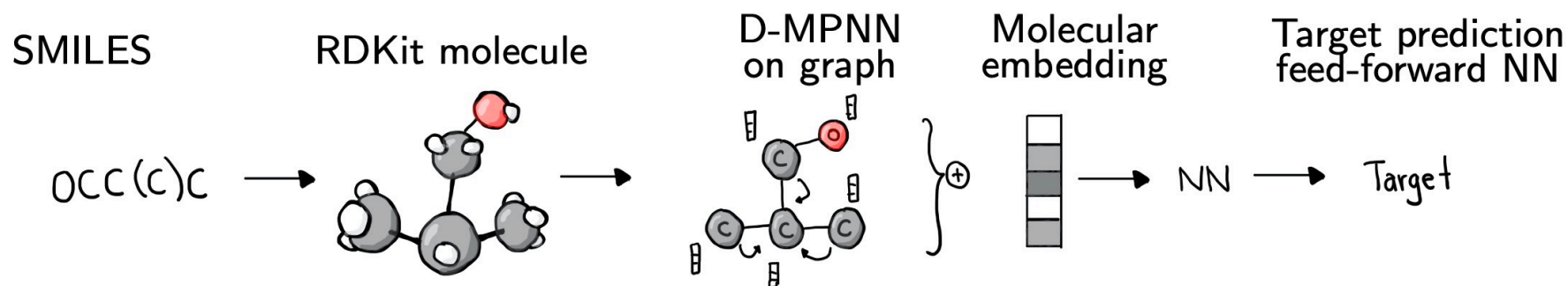
Hands-on Demo



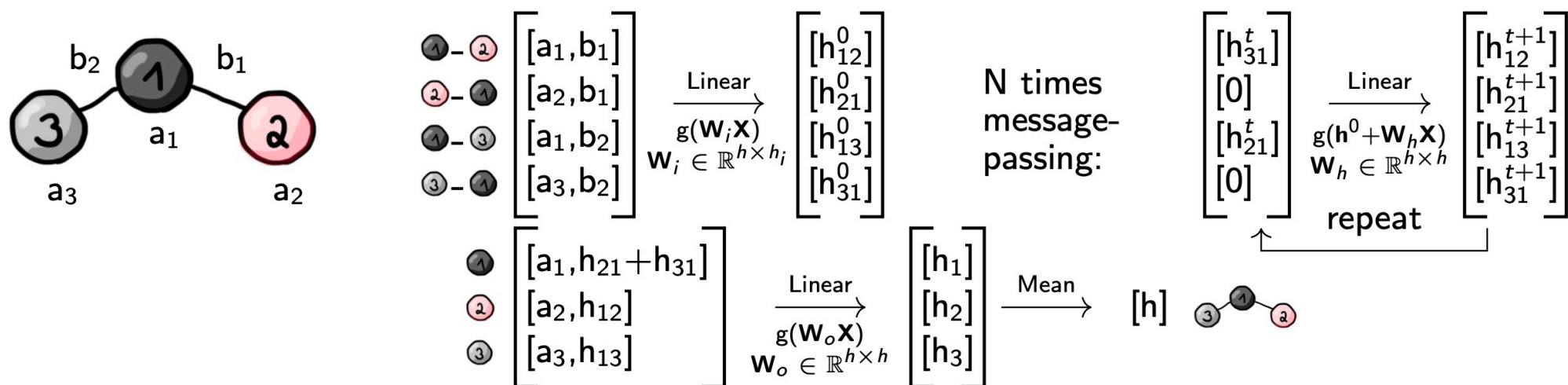
chemprop

Part B-2

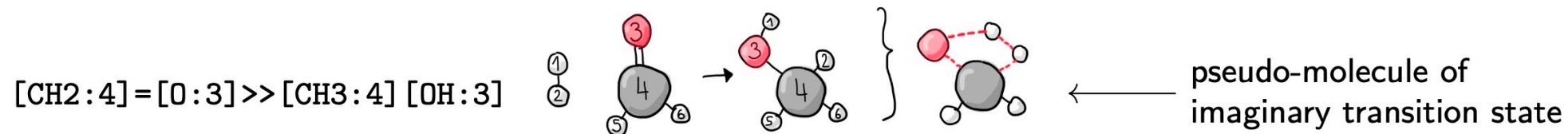
Anatomy of Chemprop (D-MPNN)



Atom (vector length 133): [Atomic num, Degree, Charge, Chirality, Hydrogens, Hybr., Aromaticity, Mass]
 Bond (vector length 14): [Bond, Bond type, Conjugation, Ring, Stereoinformation]

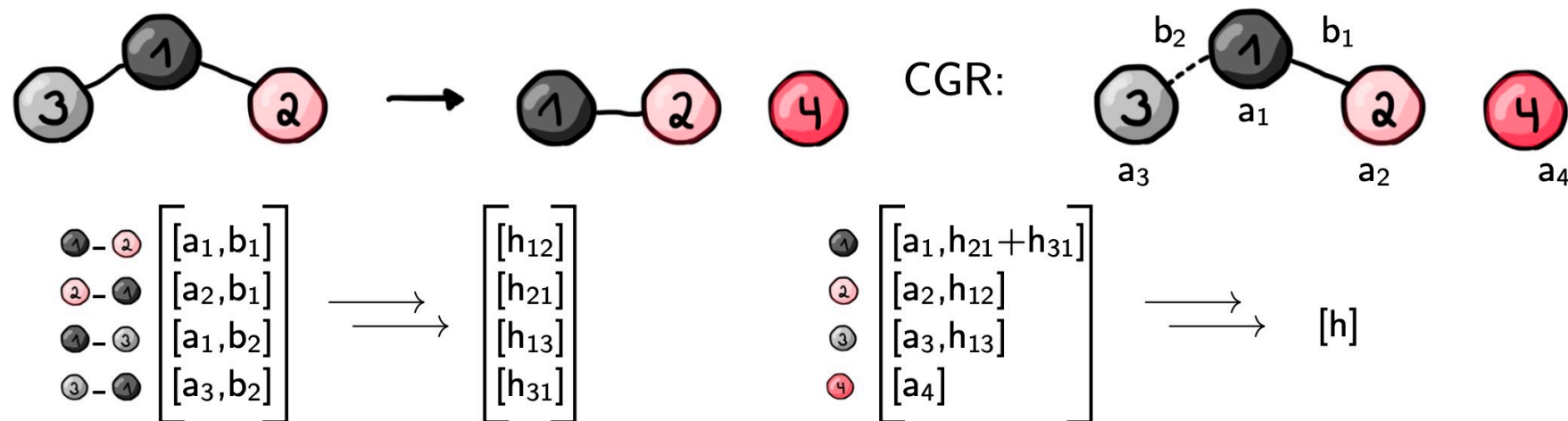


Anatomy of Chemprop (D-MPNN)



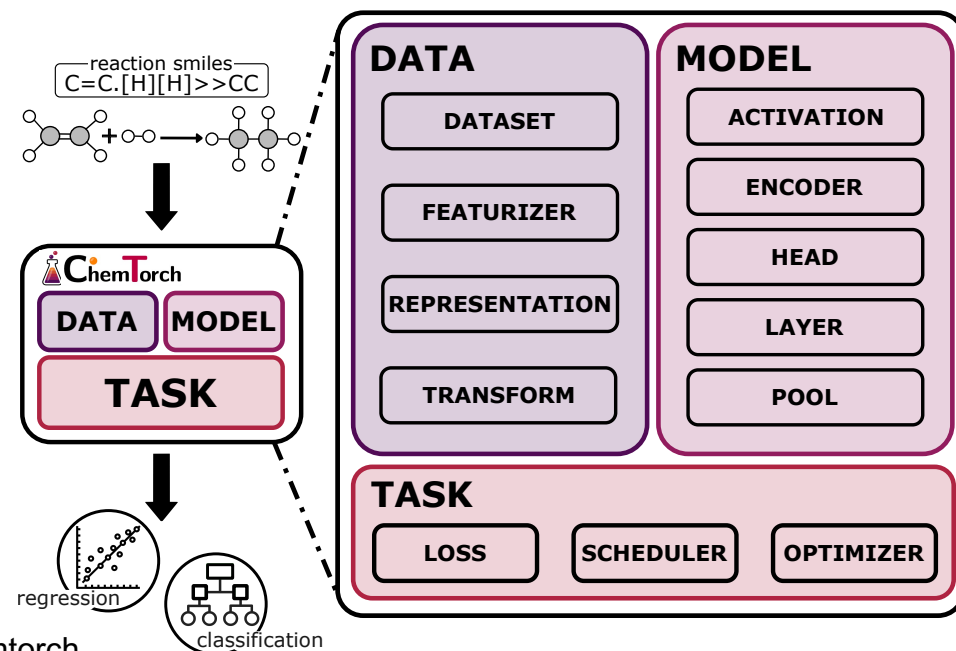
Atom (vector length 145): [At, Deg, Ch, Chir, Hydr, Hybr, Ar, M], Deg, Ch, Chir, Hydr, Hybr, Ar, M]

Bond (vector length 28): [Bond, Type, Conj, Ring, Stereo, Bond, Type, Conj, Ring, Stereo,]



Modular framework for reaction ML

- NNs on molecular/reaction fingerprints
- GCNNs on molecular/reaction graphs (D-MPNN, PNA, GAT, GCN, GatedGCN...)
- Graph transformers on molecular/reaction graphs
- Language models on molecular/reaction string representations



De Landsheere, et al., **Heid**, 2025, ChemRxiv 10.26434/chemrxiv-2025-9mggj, github.com/heid-lab/chemtorch

Attention layers

Soft dictionary lookup:

Key	Value
(1.0, 0.0)	0
(0.8, 0.5)	30
(0.9, 0.8)	50

What is the value of the query (0.7, 0.7)?

Attention: $\text{Softmax}\left(\frac{Q K^T}{\sqrt{d_K}}\right) V = \text{Softmax}\left(\frac{(0.7, 0.7) \begin{pmatrix} 1.0 & 0.8 & 0.9 \\ 0.0 & 0.5 & 0.8 \end{pmatrix}}{\sqrt{2}}\right) \begin{pmatrix} 0 \\ 30 \\ 50 \end{pmatrix} =$

Attention layers

Soft dictionary lookup:

Key	Value
(1.0, 0.0)	0
(0.8, 0.5)	30
(0.9, 0.8)	50

What is the value of the query (0.7, 0.7)?

Attention: $\text{Softmax}\left(\frac{Q K^T}{\sqrt{d_K}}\right) V = \text{Softmax}\left(\frac{(0.7, 0.7) \begin{pmatrix} 1.0 & 0.8 & 0.9 \\ 0.0 & 0.5 & 0.8 \end{pmatrix}}{\sqrt{2}}\right) \begin{pmatrix} 0 \\ 30 \\ 50 \end{pmatrix} =$

 **Similarity**

$\text{Softmax}((0.49, 0.64, 0.88)) \begin{pmatrix} 0 \\ 30 \\ 50 \end{pmatrix} =$

Attention layers

Soft dictionary lookup:

Key	Value
(1.0, 0.0)	0
(0.8, 0.5)	30
(0.9, 0.8)	50

What is the value of the query (0.7, 0.7)?

Attention: $\text{Softmax}\left(\frac{Q K^T}{\sqrt{d_K}}\right) V = \text{Softmax}\left(\frac{(0.7, 0.7) \begin{pmatrix} 1.0 & 0.8 & 0.9 \\ 0.0 & 0.5 & 0.8 \end{pmatrix}}{\sqrt{2}}\right) \begin{pmatrix} 0 \\ 30 \\ 50 \end{pmatrix} =$

Similarity

$\text{Softmax}((0.49, 0.64, 0.88)) \begin{pmatrix} 0 \\ 30 \\ 50 \end{pmatrix} =$

Probability

$(0.27, 0.32, 0.41) \begin{pmatrix} 0 \\ 30 \\ 50 \end{pmatrix} = 30.1$

Attention layers

Soft dictionary lookup:

What is the value of the query (0.7, 0.7)?

Key	Value
(1.0, 0.0)	0
(0.8, 0.5)	30
(0.9, 0.8)	50

Attention: $\text{Softmax}\left(\frac{Q K^T}{\sqrt{d_K}}\right) V$

Machine-learned attention:

Query, key and value vectors first pass through a neural network,
i.e. $Q = \text{NN}_Q(Q)$, $K = \text{NN}_K(K)$, $V = \text{NN}_V(V)$

Attention layers

Equal neighbor contributions:

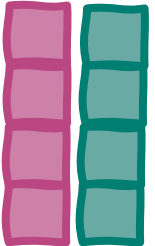
New vector  = 
+ NN()
+ NN()

Attention layers

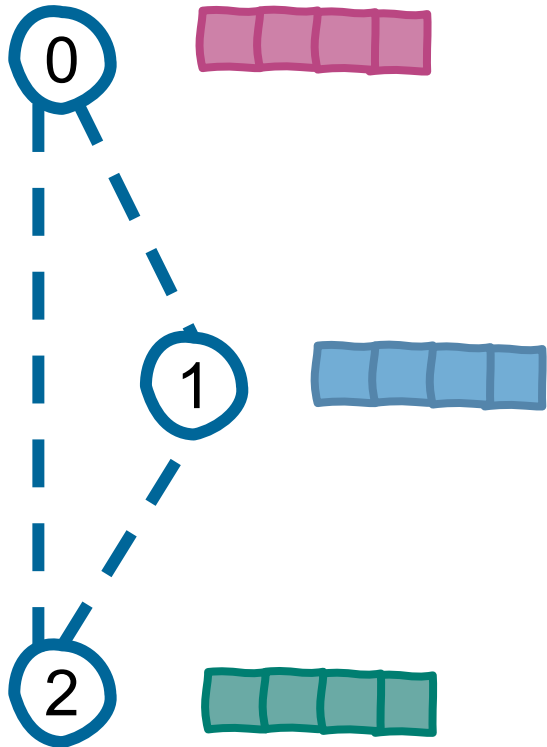
Equal neighbor contributions:

New vector  = 
+ NN()
+ NN()

Learned neighbor contributions:

 + Softmax($\frac{\text{NN}_Q(\text{blue vector}) \text{NN}_K(\text{pink/green vectors})}{\sqrt{4}}$)  

Graph transformer



New vectors

$$\begin{matrix} \text{pink vector} \\ \text{blue vector} \\ \text{green vector} \end{matrix} = \text{GlobalAttention} \left(\begin{matrix} \text{pink vector} \\ \text{blue vector} \\ \text{green vector} \end{matrix} \right)$$

... But now we need positional and structural encodings to make use of the graph structure

Hands-on Demo Part C-1 + C-2

Acknowledgements

Thank you for your attention!



Publications



Funding



Further questions: esther.heid@tuwien.ac.at

Current members:



Past members:

